

CS683 — ADVANCED COMPUTER ARCHITECTURE

Hardware-Conscious Performance Engineering

Programming Assignment 1: 2D Convolution and SGEMM

TODO Name

TODO Roll No.

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0.1 Experimental setup

Component	Configuration
CPU	13th Gen Intel Core i5-13420H 4 P-cores + 4 E-cores
L1d cache	48 KiB per P-core
L2 cache	1.25 MiB per P-core
L3 cache	12 MiB
Memory	16 GiB
Vector ISA	SSE4.2, AVX, AVX2, FMA, AVX-VNNI — no AVX-512

1 Task 1: 2D convolution

Baseline: `conv_naive`

1.1 Task 1A: Loop unrolling and reordering

1.2 Task 1B: Tiling

1.3 Task 1C: SIMD

1.4 Task 1D: Sabka saath sabka vikaas

1.5 Results

Stage	Time (ms)	GFLOP/s	Speedup vs naive
naive			1.00×
reorder			
unroll			
tile			
simd			
optimized			

Table: 2048×2048 image, $K = 3$.



Figure 1: Speedup over `conv_naive` versus image dimension, at fixed $K = 3$.



Figure 2: Speedup over `conv_naive` versus kernel size K , at a fixed image size.

1.6 Discussion

2 Task 2: Matrix multiplication (SGEMM)

Baseline: `matmul_naive`

2.1 Task 2A: Software prefetching

2.2 Task 2B: SIMD (Single Instructor Multiple Deadlines)

2.3 Task 2C: Software prefetching + SIMD

2.4 Results

Stage	Time (ms)	GFLOP/s	Speedup vs naive
naive			1.00×
simd			
prefetch			
optimized			

Table: $M = N = K = 1024$.

plot pending

Figure 3: Speedup over `matmul_naive` versus matrix size, $M = N = K$ from 128 to 2048.

plot pending

Figure 4: Speedup of `matmul_prefetch` versus prefetch distance.

plot pending

Figure 5: Speedup of `matmul_prefetch` for the `_MM_HINT_T0` / `T1` / `T2` / `NTA` locality hints.



plot pending

Figure 6: Speedup versus SIMD width: 128-bit SSE, 256-bit AVX2, 512-bit AVX-512.

2.5 Integration with llama.cpp

2.6 Discussion

3 Conclusion